

# Topic 1.14 - Function Model Construction and Application

## Guided Notes

Strengthened ECHS edition - reasoning, challenge, and AP-style practice

Name: \_\_\_\_\_

Class: \_\_\_\_\_

Date: \_\_\_\_\_

### Why this topic matters

Constructing a model is not only fitting an equation. It requires choosing a structure, using conditions to determine parameters, checking that the model satisfies those conditions, applying it to answer a question, interpreting rates and values with units, and evaluating whether the result is meaningful in context.

### Learning targets

- Construct linear, quadratic, polynomial, rational, and piecewise models from stated conditions or data.
- Use models to solve equations, compare rates, and interpret parameters and outputs in context.
- Validate a model against conditions, domain, residual evidence, assumptions, and practical constraints.

### Language and structure

| Term                     | Meaning in this topic                                                               |
|--------------------------|-------------------------------------------------------------------------------------|
| <b>parameter</b>         | A constant in a model whose value controls a structural feature.                    |
| <b>calibration</b>       | Determining parameter values from data or conditions.                               |
| <b>validation</b>        | Checking a model against observations or requirements not merely used to create it. |
| <b>piecewise model</b>   | A model using different rules on different parts of the domain.                     |
| <b>contextual domain</b> | The input values for which the model and its assumptions are meaningful.            |

### Launch - reason before calculating

A parabola has vertex  $(3, 20)$  and passes through  $(0, 2)$ . Construct its equation and explain why the information determines a unique quadratic in vertex form.

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## Concept 1: Constructing from geometric and algebraic conditions

### Core idea

Choose a form aligned to the information: slope-intercept for slope and intercept, vertex form for a quadratic vertex, factored form for zeros, and transformed rational form for asymptotes. Substitute remaining points to determine scale, then verify every condition.

### Worked example - annotate the reasoning

**Problem.** Construct the quadratic with vertex  $(3, 20)$  through  $(0, 2)$ .

**Model reasoning.**  $y = a(x - 3)^2 + 20$  and  $2 = 9a + 20$ , so  $a = -2$ . The model is  $y = -2(x - 3)^2 + 20$ .

### Check your understanding

Independent

No calculator

Construct the quadratic with vertex  $(-2, 5)$  through  $(0, 13)$ .

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## Concept 2: Calibrating a rational model from asymptotes and values

### Core idea

A quantity approaching a long-run level  $L$  can often be modeled by  $M(t) = L + A/(t + b)$ . The horizontal asymptote fixes  $L$ ; two observed values can determine  $A$  and  $b$ . Restrictions and the contextual domain must be stated separately.

### Worked example - annotate the reasoning

**Problem.** A model approaches 80, with  $M(0) = 180$  and  $M(5) = 130$ . Find  $M(t) = 80 + A/(t + b)$ .

**Model reasoning.**  $A/b = 100$ , so  $A = 100b$ . Also  $A/(b + 5) = 50$ , so  $A = 50b + 250$ . Hence  $b = 5$ ,  $A = 500$ , and  $M = 80 + 500/(t + 5)$ .

### Check your understanding

Independent

No calculator

A model approaches 20, with  $N(0) = 70$  and  $N(5) = 45$ . Determine  $N(t) = 20 + A/(t + b)$ .

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### Concept 3: Piecewise construction, application, and validation

#### Core idea

Piecewise models encode changes in rules. Define each interval clearly, check endpoints and continuity when appropriate, and use the correct branch. After solving, interpret units and test whether the answer lies in the branch and contextual domain.

#### Worked example - annotate the reasoning

**Problem.** A parking garage charges 8 riyals for up to 3 hours and then 2.5 riyals per additional hour. Construct a continuous model for  $0 \leq h \leq 12$ .

**Model reasoning.**  $C(h) = 8$  for  $0 \leq h \leq 3$  and  $C(h) = 8 + 2.5(h - 3)$  for  $3 < h \leq 12$ . Both branches agree at  $h = 3$ , so the model is continuous there.

#### Check your understanding

Independent

No calculator

Use the model to find the charge for 7.5 hours and solve for the time producing a charge of 25.50.

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### Representation laboratory

#### Connect at least two representations

A bridge arch passes through  $(0, 0)$ ,  $(5, 10)$ ,  $(10, 0)$ . Construct the model using both factored and vertex forms, and explain what each form makes visible.

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## Challenge laboratory

### Challenge - justify every claim

Construct a least-degree real polynomial with zeros  $-2$  simple and  $3$  double, with  $P(0) = 36$  and right end down. State the degree, leading coefficient, and a reasonable sign chart.

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## Error clinic and calculator strategy

### Common errors to diagnose

- Using a convenient form that does not encode the supplied features efficiently.
- Solving for a parameter but failing to verify all original conditions.
- Reporting an algebraic solution that lies outside the branch or contextual domain.
- Treating a fitted equation as exact law without checking residuals or assumptions.

### Technology decision

Regression can determine parameters from many data points, but report the model family, window, residual evidence, and rounding policy. Keep full calculator precision internally, validate against data, and interpret only to a precision supported by the measurements.

**AP-style synthesis****Multi-part reasoning task**

A pedestrian arch touches the ground at horizontal positions  $x = 0$  and  $x = 10$  and reaches height 10 at  $x = 5$ .

**A.** Construct a quadratic height model in factored form.

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**B.** Rewrite the model in vertex form and identify the maximum.

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**C.** Find the average rate of change from  $x = 2$  to  $x = 8$  and explain why the value can be zero although the arch is not horizontal throughout.

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**Exit ticket****Before you leave**

1. Which quadratic form is best when zeros are given?

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2. What must be checked after solving with a piecewise model?

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3. Why should regression coefficients not be rounded too early?

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